BioinfoAgentBench: Problem-Structure-Centered Benchmark for Black-Box Bioinformatics AI Agents

BFF Paradigm-Baseline·Fingerprint·Frontier

Data Curation
(3,238 tasks + 216 challenges)

Baseline Tier



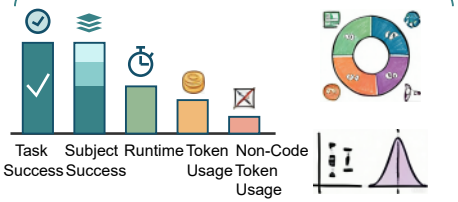
3,238
public tasks

Task Curation & Standardization

- 486 deduplicated public tasks
- Subject description(DNA, scRNA, Genetic, etc.)
- Unified format: (1) Natural language description. (2) Input data, etc.



Black-box execution

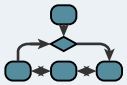


Fingerprint Tier

Planning(P)

P1, Workflow Design Load

P2, Branch Decision Load

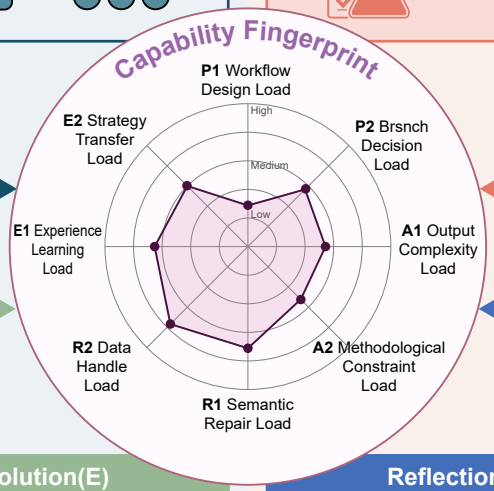


Action(A)

A1, Output Complexity Load

A2, Methodological Constraint Load





Evolution(E)

E1, Experience Learning Load

E2, Strategy Transfer Load



Reflection(R)

R1, Semantic Repair Load

R2, Data Handle Load



Frontier Tier



216 Real-world
Scienific Challenges



Jupyter Notebook
Reference Reconstruction

- Reproduction gap
- Reasoning gap
- Insight gap



Seen(80%)
Unseen(20%)

Discovery-gap Evaluation



Black-Box Evaluation Protocol → Cross-system Capability Fingerprint → Webserver for Community Access

